

Xinyu Xie
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Mississippi State University
Starkville, US

Research Interests

- How task representation affects strategy selection in problem solving
- How strategy selection associates with individual differences
- The underlying cognitive mechanisms and practical consequences of strategy switching

Education

Mississippi State University Ph.D. in Psychology, Cognitive Science M.S. in Psychology Advisor: Dr. Jarrod Moss	Starkville, USA 2021 – Present 2021 – 2024
Beijing Forestry University B.S. in Applied Psychology	Beijing, China 2016 – 2020

Publications

Peer-reviewed Journal Papers

Xie, X., & Moss, J. (2025). Task representation and individual differences affect strategy selection and problem-solving performance. *Frontiers in Psychology*, 16.

Peer-reviewed Proceedings Papers

Xie, X., & Moss, J. (2024). Comparing theories that posit a role for task features in strategy selection. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 46(0).

Conference Presentations

Xie, X., & Moss, J. (2024). *Comparing theories that posit a role for task features in strategy selection*. Proceedings of the 46th Annual Conference of the Cognitive Science Society, Rotterdam, Netherlands.

Xie, X., & Moss, J. (2023). *Interaction between recency effect and strategy preferences in problem solving*. Psychonomic Society 64th Annual Meeting, San Francisco, California, USA.

Xie, X., & Moss, J. (2022). *Working memory capacity predicts more effective problem space exploration*. Proceedings of the 44th Annual Conference of the Cognitive Science Society, Toronto, Canada.

Teaching Experience

2025 Spring Instructor, Cognitive Psychology (PSY 3713-01, size: 50)

2025 Fall Instructor, Cognitive Psychology (PSY 3713-03, size: 50)

Honors Awards

2025 Peyman Psychology Graduate Student Scholarship

2024	Travel Assistance Grant for Graduate Students (TAGGS) (2 students per year)
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2022 GRAG Fellowship

2021 Peyman Psychology Graduate Student Scholarship

Research Experience

Research Assistant, Problem Solving and Learning Lab

08/2021 – Present

Advised by Prof. Jarrod Moss, Mississippi State University

Starkville, USA

- Designed and conducted a study to explore the influence of task representation and individual differences in working memory capacity, attentional control, and inductive reasoning on strategy selection in a problem-solving task; collected 144 participants data in total
- Designed and conducted a study to compare two theories' predictions about how task representation affects strategy selection in a problem-solving task; collected around 430 participants data in total
- Prepared and submitted 1 peer-reviewed manuscripts, 1 peer-reviewed conference paper, and 2 conference abstracts
- Developed skills related to experimental research design, data analysis, academic writing, and research presentation

Research Assistant, Social Cognition and Neuroeconomics Lab

04/2019 – 05/2019

Advised by Prof. Tingyong Feng, Southwest University

Chongqing, China

- Assisted with the study entitled "The preference for future outcomes correlates with the temporal variability of functional connectivity among brain regions" to complete the transcranial direct current stimulator (tDCS) experiment; collected 20 participants data
- Collected and encoded questionnaires (about 100 participants' data); analyzed the encoding data for the study entitled "Neural substrates underlying episodic future thinking: A voxel-based morphometry study"
- Obtained hands-on experience in tDCS; studied the data processing of resting-state fMRI, conducted data pre-processing and visualization

National Innovation and Entrepreneurship Program: The effects of aerobix exercise in forest on the executive functions of college students with internet addiction 04/2018 – 05/2019

04/2018 – 05/2019

Advised by Prof. Shiyue Sun, Beijing Forestry University

Beijing, China

- Designed the experiment by using E-prime; Conducted a pilot study and formal study; collected 30 participants data
- Completed data analysis and manuscript writing; Accepted by Academic Annual Meeting of Chinese Association of Social Psychology in 2019

Undergraduate Thesis: The impact of episodic future thinking on intertemporal decision

Advised by Prof. Yang Yang, Beijing Forestry University

Beijing, China

- Designed questionnaires to conduct preliminary investigation of experimental materials to ensure the reliability and validity of the experimental materials
- Developed an E-prime program for the intertemporal decision task; collected 30 participants data in total
- Completed data analysis, thesis writing, and defense (score: 90/100, top 5%)

Internship Experience

Schneider-Electric (China) Ltd. R&D Center

08/2020 – 01/2021

Talent Acquisition & Mobility Assistant

Shanghai, China

- Assisted with interview arrangements (more than 70 candidates a week), including time and room coordination, and candidate reception, etc.
- Participated in global campus hiring project, responsible for 5 headcounts campus hiring independently

Academic Advising

Undergraduate

2024 - 2025 Maggie Freeman, Mississippi State University

2023 - 2024 Ty Smith, Mississippi State University

Service Outreach

Reviewer

2025 Cognitive Science Conference

2024 Cognitive Science Conference

Judge

2025 Undergraduate Research Symposium

Quantitative Skills

Programming Python, R

Others SPSS, Linux, L^AT_EX